

- The radiation requirements for Total Ionizing Dose (**TID**), Non-Ionizing Energy Losses (**NIEL**) and Single Event Effects (**SEE**) were established by the voxel with the highest simulated irradiation value, affecting 128 out of the 896 DB positions (**Fig. 4a, 4b and 4c**).
- During production phase, 10 samples of each batch of the active components acquired were tested to account for possible batch differences (except for big/highly dense components like FPGAs)
- **Safety Factors** were applied to account for inconsistencies in **simulation (1.5)** and in tests with **mono-energetic beams (1.3)**
- The requirements of: **TID: 32.71 Gy**, **NIEL: 4.38×10^{12} (1 MeV n)/cm²**, and **SEE: 2.37×10^{11} h/cm²**.
- The components were pre-selected and tested with higher dose and fluence to account for lot variation [4, 5].

